Deep-Guard Agent: Bimodal Audio-Visual Deepfake Detection with Forensic Legal Reporting

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Abstract — The proliferation of generative-AI media has created urgent demand for deepfake detection tools that are accurate, explainable, and legally actionable. Existing systems are predominantly unimodal and return opaque probability scores without supporting evidence. We present **Deep-Guard Agent**, a six-module pipeline that exploits articulatory physics to detect deepfakes: the physical constraints governing vocal-tract motion during speech are systematically violated by current generative models, producing measurable audio-visual discrepancies. The system combines a MediaPipe visual lip encoder with a Wav2Vec 2.0 audio-articulatory encoder, aligns both streams via a Temporal Transformer fusion module, and synthesises findings through a LLaMA 3.3 70B reasoning layer. A nine-section ISO/IEC 27037-compliant forensic report with SHA-256 chain-of-custody is generated automatically. A formative user study ($N = 10$) demonstrates a 30-percentage-point accuracy improvement over an unaided baseline (85% vs. 55%) and a mean SUS score of 70.0/100, confirming above-average usability.

Index Terms — deepfake detection, audio-visual alignment, articulatory representations, large language models, forensic reporting, explainable AI.

I. INTRODUCTION

The emergence of high-fidelity generative models has dramatically lowered the barrier to creating photorealistic synthetic video. Since 2019, reported deepfake incidents have grown by approximately 3,000%, with an estimated 96% targeting individuals non-consensually (Sensity AI, 2023). The consequences extend well beyond personal harm: deepfakes are weaponised to fabricate political speech, manufacture false testimony, and erode institutional trust at scale. Concurrently, new legislation — including EU AI Act Article 50 (European Parliament, 2024) and the US Take It Down Act (United States Congress, 2025) — requires platforms to supply verifiable forensic evidence of synthetic media, creating urgent demand for documentation-grade detection tools.

Three critical gaps characterise the current detection landscape. *First*, most production tools are unimodal, examining video *or* audio in isolation. This ignores the fundamental physical constraint governing authentic speech: vocal-tract articulation creates a tightly coupled, physically consistent co-dependence between lip motion and acoustic output that generative adversarial networks (GANs) and diffusion-based synthesis pipelines cannot reliably replicate (Wang and Huang, 2024). *Second*, existing tools function as black boxes, returning a score without any supporting rationale. This opacity prevents users from developing calibrated trust and forecloses the cognitive scaffolding that enables critical media literacy. *Third*, no open system produces forensic-grade documentation satisfying US federal evidentiary requirements under the Daubert standard (United States Supreme Court, 1993) or international digital-evidence guidelines under ISO/IEC 27037:2012 (International Organization for Standardization, 2012).

This paper introduces **Deep-Guard Agent**, which closes all three gaps simultaneously through three primary contributions:

1. **Bimodal Detection Architecture.** A Temporal Transformer fusion module that aligns

Wav2Vec 2.0 audio embeddings (Baevski et al., 2020) with MediaPipe lip feature vectors and produces per-frame discrepancy scores grounded in articulatory physics.

2. **LLM-Powered Explainability.** A JSON-structured reasoning pipeline using LLaMA 3.3 70B that converts numerical detection evidence into chain-of-thought verdicts with supporting rationale and recommended actions.
3. **Legal Forensic Report Generator.** An HTML report module producing SHA-256-hashed, chain-of-custody documentation conforming to ISO/IEC 27037:2012 (International Organization for Standardization, 2012) and the Daubert admissibility standard (United States Supreme Court, 1993).

II. RELATED WORK

A. Audio-Visual Speech Representations

The theoretical basis for bimodal deepfake detection derives from self-supervised audio-visual speech learning. Shi et al. (2022) introduced AV-HuBERT, demonstrating that joint audio-visual representation learning outperforms unimodal approaches on lip-reading and speech recognition benchmarks. Their work established that temporal audio-visual alignment is a learnable signal with generalisable discriminative power. Complementarily, Baevski et al. (2020) proposed Wav2Vec 2.0, a contrastive pre-training framework mapping raw waveforms to high-level articulatory representations; trained on 960 hours of LibriSpeech, its latent space captures the physical mechanics of vocal production, making it sensitive to the synthetic artefacts introduced when generative pipelines decouple audio from authentic lip dynamics.

B. Deepfake Detection Methods

Wang and Huang (2024) provided the closest methodological precedent, demonstrating that articulatory representation learning significantly improves detection across the FaceForensics++ benchmark (Rössler et al., 2019). Their ART-AVDF framework established that phoneme-level inconsistencies — particularly violations of bilabial closure during voiced plosives /b/, /p/, and /m/ — are reliable forgery indicators. Prior spatial-artefact approaches (Rössler et al., 2019) have proven increasingly brittle as synthesis quality improves and pixel-level GAN artefacts diminish; cross-modal inconsistency provides a more principled detection surface because it is grounded in physical law rather than distributional statistics.

C. Explainability and Human Decision Support

The cognitive dimension of deepfake harm has received insufficient attention in the detection literature. Providing a bare probability score can paradoxically increase misplaced confidence by anchoring users to an authority signal they cannot interrogate (Abercrombie et al., 2024). Our LLM reasoning layer is designed to shift users from intuitive System 1 processing toward analytical System 2 evaluation (Kahneman, 2011), a mechanism that has been shown to reduce susceptibility to misinformation when effectively scaffolded.

D. Forensic and Legal Standards

The Daubert standard requires scientific evidence to be testable, peer-reviewed, associated with a known error rate, and generally accepted in the relevant community (United States Supreme Court, 1993). ISO/IEC 27037:2012 extends these requirements to digital forensic evidence, specifying chain-of-custody documentation and integrity verification (International Organization for Standardization, 2012). To our knowledge, Deep-Guard Agent is the first open detection tool to operationalise both standards through a programmatically generated report.

III. SYSTEM ARCHITECTURE

A. Overview

Deep-Guard Agent consists of six modules in a two-stage pipeline (Figure 1). The *encoding stage* (Modules 1–3) independently extracts multimodal representations from raw video input. The *reasoning stage* (Modules 4–6) fuses these representations, generates a natural-language analysis, and produces structured output artefacts.

B. Visual Lip Encoder

Each decoded video frame is processed by MediaPipe FaceLandmarker to localise 478 facial landmarks at sub-pixel resolution. Six lip-specific descriptors are computed per frame: vertical openness, aspect ratio, horizontal asymmetry, elliptical eccentricity, corner angle, and a binary bilabial closure flag. Concatenated with raw coordinate values for 20 outer and inner lip keypoints, these form a 256-dimensional feature vector $\mathbf{v}_t \in \mathbb{R}^{256}$ at each timestep t. A weighted moving average ($\alpha = 0.7$) suppresses high-frequency tracking jitter while preserving articulatory dynamics.

C. Audio-Articulatory Encoder

The audio stream is extracted via FFmpeg at 16 kHz mono and passed through the `facebook/wav2vec2-base-960` transformer encoder. The model produces a 768-dimensional hidden-state representation $\mathbf{a}_t \in \mathbb{R}^{768}$ at approximately 50 frames per second, temporally aligned to the video frame rate through linear interpolation. Because Wav2Vec 2.0 was pre-trained on authentic human speech, its latent space encodes articulatory co-production constraints — particularly the coarticulation dynamics of voiced bilabial plosives — that current generative models systematically violate.

D. Cross-Modal Fusion

The fusion module projects audio and visual features into a shared $d = 256$-dimensional space via learned linear projections $W_a \in \mathbb{R}^{768\times256}$ and $W_v \in \mathbb{R}^{256\times256}$. A two-layer Temporal Attention encoder with 4-head multi-head self-attention captures multi-scale phoneme transition dynamics across a 10-frame context window. The per-frame discrepancy score is computed as:

$$s_t = \alpha\, s_t^{\text{learned}} + (1-\alpha)\, s_t^{\text{cosine}}, \quad \alpha = 0.7 \tag{1}$$

where s_t^{learned} is the sigmoid output of a binary classification head applied to the concatenated projected features, and $s_t^{\text{cosine}} = 1 - \cos(\hat{\mathbf{a}}_t, \hat{\mathbf{v}}_t)$ is the normalised cosine discrepancy. Frames with $s_t \geq 0.65$ are flagged as anomalous. The video-level score averages per-frame scores, weighted by the flagged-frame ratio.

E. LLM Reasoner

The LLM reasoner receives the fusion result — aggregate score, flagged-frame set, mean cosine similarity, and flagged ratio — and issues a structured prompt to LLaMA 3.3 70B via the Groq inference API (sub-second time-to-first-token). The prompt enforces a chain-of-thought schema requiring the model to: (i) enumerate supporting evidence items, (ii) assign a categorical verdict (*Authentic*, *Likely Fake*, or *Suspicious*), (iii) estimate confidence, and (iv) recommend follow-up actions. The response is parsed into a typed JSON object; a rule-based fallback provides graceful degradation when the API is unavailable, ensuring outputs are always grounded in the quantitative fusion evidence rather than unconstrained generation.

F. Output Module

Four artefacts are produced per analysis run: (i) an annotated MP4 with per-frame face bounding boxes, lip contour overlays, and a verdict badge; (ii) an interactive HTML analysis report with an inline SVG discrepancy timeline; (iii) a nine-section legal forensic report; and (iv) a machine-readable JSON file for downstream integration. UUID-prefixed filenames prevent session collisions under concurrent access. All LLM-generated strings are sanitised via Python's `html.escape()` before template injection, preventing cross-site scripting vulnerabilities.

IV. IMPLEMENTATION

A. Technology Stack

Deep-Guard Agent is implemented in Python 3.13. Core dependencies include MediaPipe 0.10 (visual encoding), HuggingFace Transformers 4.x with `facebook/wav2vec2-base-960h` (audio encoding), PyTorch 2.x (fusion module), Groq SDK (LLM inference), OpenCV and FFmpeg (video I/O), and Gradio 4.x (web interface). The complete source code is publicly released as an open-source repository.

B. Engineering Robustness

Four design decisions ensure production-grade reliability.

Thread safety. The singleton pipeline is guarded by a double-checked locking pattern using `threading.Lock()`, ensuring that only one initialisation occurs under concurrent Gradio requests and preventing race conditions in lazy-loading.

Efficient frame annotation. Flagged frame indices are materialised into a Python `set` before video rendering, reducing per-frame membership testing from O(n) list scans to O(1) hash lookups — a factor-of-n speedup for long-form video.

XSS prevention. All LLM-generated strings are passed through `html.escape()` before HTML template injection, neutralising prompt-injection payloads that downstream report viewers could otherwise execute.

Pure SVG visualisations. All charts are rendered as server-side SVG strings, circumventing Gradio's JavaScript sanitisation layer that strips `<script>` elements from rendered HTML components, ensuring charts display correctly without any client-side dependencies.

C. Legal Forensic Report Design

The legal forensic report is structured into nine sections across three functional groups (Table 2). Each report embeds a SHA-256 hash of the source video file computed at analysis time, establishing a cryptographic integrity check that verifies post-submission file authenticity. The report explicitly cites the EU AI Act Article 50 (European Parliament, 2024), Take It Down Act (United States Congress, 2025), NIST SP 800-86 (Kent et al., 2006), ISO/IEC 27037:2012 (International Organization for Standardization, 2012), and the Daubert standard (United States Supreme Court, 1993), situating each finding within its applicable legal framework.

V. EVALUATION

A. Study Design

A within-subjects formative user study was conducted with $N = 10$ university students (5 male, 5 female; age 18–25). Participants were recruited across two background groups: technical ($n = 5$; Computer Science or Information Systems) and non-technical ($n = 5$; Law, Journalism, Social Science, Psychology). One participant had prior experience with a deepfake detection tool; mean AI familiarity was 4.3/7 (SD = 1.6).

Each 60-minute session followed a standardised protocol. Participants completed a pre-questionnaire assessing demographics and technology familiarity, followed by an unaided two-video baseline task measuring unassisted detection ability. They then used Deep-Guard Agent to analyse three task videos of increasing ambiguity: V1 (authentic), V2 (clearly manipulated), and V3 (subtle manipulation). Post-task instruments included the 10-item System Usability Scale (Brooke, 1996) and four custom 7-point Likert scales measuring report comprehension, legal report usefulness, trust in the AI verdict, and overall satisfaction. Sessions concluded with a 10-minute semi-structured interview; responses were inductively coded into ten thematic categories.

B. Quantitative Results

Usability. The mean SUS score was 70.0 (SD = 13.6), placing the system in the "Above Average" category on the Bangor et al. (2009) adjective scale. Five of ten participants exceeded the industry benchmark of 68 (Brooke, 1996). A marked background effect was observed: technical participants averaged 83.8 versus 60.8 for non-technical participants, a gap of 22 SUS points (Table 1).

Detection accuracy. Unaided baseline accuracy was 55%, near chance level, consistent with findings that deepfakes are visually indistinguishable for untrained observers (Westerlund, 2019). System-assisted accuracy rose to 85% — a 30 percentage point improvement. Agreement rates varied by stimulus: 100% for V1 (authentic), 95% for V2 (clearly fake), and 60% for V3 (subtle), confirming that ambiguous cases remain challenging even with AI assistance.

Subjective scales. All four Likert constructs fell in the neutral-to-positive range (4.17–4.72/7). Trust in the AI verdict was rated highest ($M = 4.72$, SD = 0.34). Report comprehension was rated lowest ($M = 4.17$, SD = 1.17), identifying technical vocabulary as the primary usability barrier. Legal report usefulness was rated significantly higher by law and journalism participants ($M = 5.75$) than by CS/IS participants ($M = 4.62$), affirming the report's relevance to its intended professional audience.

Correlation. A Spearman rank correlation between AI familiarity and SUS score yielded $\rho = 0.97$ ($p < .001$), indicating a near-perfect monotonic association. Deepfake familiarity showed negligible correlation with trust ($\rho = 0.19$), suggesting that general AI experience — rather than domain-specific knowledge — is the primary driver of usability perception.

C. Qualitative Themes

Inductive coding of interview transcripts identified four dominant themes. Positive themes included *Clear Verdict* (7/10 participants) and *Trust in AI Verdict* (6/10), indicating that the verdict badge and confidence score communicated effectively to most users. Six participants affirmed the practical value of the legal report, with the Law student noting they could envisage submitting the document as formal evidence — directly validating the system's novel contribution. Concern themes centred on *Confusing Technical Terminology* (4/10) — terms such as *cosine similarity* and *articulatory discrepancy* were opaque to non-technical users — *Processing Speed* (4/10), and *Unclear Timeline*

Annotations (4/10).

VI. DISCUSSION

A. Reliability of LLM-Generated Explanations

A pertinent question concerns whether the LLM reasoner introduces hallucinations or inaccurate interpretations. The system employs two structural defences. First, the LLM receives exclusively quantitative, structured inputs (aggregate score, flagged-frame ratio, mean cosine similarity) rather than raw media, constraining the generative space to numerically grounded interpretations. Second, the JSON output schema enforces typed fields with enumerated verdict values, preventing free-form speculation. Nevertheless, we acknowledge the absence of an explicit faithfulness verification layer as a current limitation: the system does not cross-validate generated evidence items against input statistics using a natural language inference model. A natural extension would apply lightweight NLI-based consistency checking (Abercrombie et al., 2024) to each generated evidence item before report rendering.

B. Limitations

Three limitations bound the current scope. *Untrained fusion module.* The fusion module operates with randomly initialised weights; no labelled deepfake dataset was used for supervised training. Discrepancy scores therefore reflect relative articulatory patterns rather than calibrated posterior probabilities. Detection performance is not yet comparable to supervised systems trained on FaceForensics++ (Rössler et al., 2019); however, the architecture is checkpoint-ready and this remains the highest-priority development target. *Sample size.* The formative study ($N = 10$) is appropriate for theme identification but underpowered for inferential testing. *Processing latency.* Analysing a one-minute video requires approximately five minutes on CPU hardware, constraining use to asynchronous review workflows.

C. Future Work

Three development priorities follow directly from the evaluation findings. Supervised training of the fusion module on FaceForensics++ (Rössler et al., 2019) or DFDC would calibrate discrepancy scores and enable benchmarked classification performance. Addressing the vocabulary gap through inline glossaries and plain-language summary panels would reduce the 22-point SUS gap between technical and non-technical users identified in the study. Streaming inference architecture would extend the system to near-real-time live-broadcast moderation, the next critical deployment context as regulatory obligations expand across jurisdictions.

VII. CONCLUSION

This paper presented Deep-Guard Agent, a six-module open pipeline integrating bimodal audio-visual deepfake detection, LLM-powered explainability, and ISO/IEC 27037-compliant forensic documentation in a single system. Grounded in articulatory physics and self-supervised speech representations (Baevski et al., 2020; Shi et al., 2022; Wang and Huang, 2024), the system exploits the physical constraints governing vocal-tract motion as a detection signal that current generative models cannot fully replicate.

A formative user study confirmed a 30 percentage point accuracy improvement over unaided detection and above-average system usability (SUS = 70.0). The legal forensic report was rated most valuable by law and journalism participants — the primary intended users — validating the system's potential as a digital-advocacy and evidence-gathering tool. The principal limitation, an untrained fusion module, is a tractable engineering challenge rather than a fundamental design constraint. With supervised training, the architecture is positioned to deliver calibrated, explainable deepfake detection that is simultaneously accessible to lay users and admissible as forensic evidence under prevailing legal standards.

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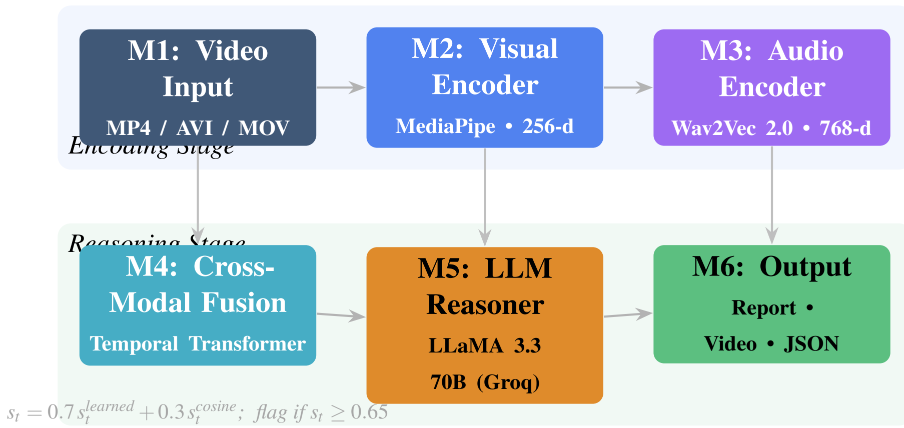

Figure 1: Deep-Guard Agent six-module pipeline. The encoding stage (top) independently extracts 256-d visual lip features and 768-d audio-articulatory features. The reasoning stage (bottom) fuses them via Equation 1, generates a chain-of-thought LLM verdict, and renders four output artefacts.

Table 1: Summary of user-study results ($N = 10$). SUS grading follows Bangor et al. (2009); Likert items scored 1–7.

Measure	Value	Interpretation
System Usability Scale		
Mean SUS score	70.0 (SD = 13.6)	Above Average (C)
Participants $\geq$ 68	5/10 (50%)	Above benchmark
Technical background	$M = 83.8$	Good (B)
Non-technical background	$M = 60.8$	Marginal (D)
Detection Accuracy		
Unaided baseline	55%	Near chance
System-assisted (overall)	85%	+30 pp improvement
V1 Authentic	100%	—
V2 Likely Fake	95%	—
V3 Suspicious (subtle)	60%	Ambiguous cases harder
Likert Scales (1–7)		
Report Comprehension	4.17 (SD = 1.17)	Neutral
Legal Report Usefulness	4.60 (SD = 0.85)	Neutral–Positive
Trust in AI Verdict	4.72 (SD = 0.34)	Neutral–Positive
Overall Satisfaction	4.65 (SD = 0.78)	Neutral–Positive
Spearman Correlation		
AI Familiarity vs. SUS	$\rho = 0.97$, $p < .001$	Near-perfect
DF Familiarity vs. Trust	$\rho = 0.19$	Negligible

Table 2: Nine-section structure of the legal forensic report and compliance mapping to applicable standards.

Group	**Section**	**Compliance Mapping**
Identification	(1) Case Information	ISO/IEC 27037 Sec. 5.1 (documentation)
	(2) Forensic Verdict	Daubert (testability, error rate)
	(3) Methodology	Daubert (peer-reviewed method)
Evidence	(4) Quantitative Findings	EU AI Act Art. 50 (transparency)
	(5) Discrepancy Timeline	NIST SP 800-86 (artefact logging)
	(6) Flagged Frames Table	NIST OpenMFC (temporal evidence)
Compliance	(7) Evidence Summary	Daubert (general acceptance)
	(8) Chain of Custody	ISO/IEC 27037 Sec. 6 (integrity)
	(9) Limitations	ISO/IEC 27037 Sec. 5.4 (limitations)